



Problem-Solution Fit

Date	27 June 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine.
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

The Problem–Solution Fit Canvas helps evaluate how effectively the proposed solution addresses the needs of the target users. It identifies customer problems, root causes, behaviors, limitations, and existing solutions, ensuring that the Emotion Detection & Learning Support Engine delivers personalized, practical, and impactful learning support.

1. CUSTOMER SEGMENT(S) [CS] - Students facing learning difficulties - Students experiencing stress or anxiety - Teachers and mentors	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited study time - Internet availability - Lack of emotional support - Different learning styles	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Pros - Online learning platforms - AI chatbots - Study forums Cons - No emotion awareness - Generic responses - Limited personalization
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2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Difficulty understanding study materials • Lack of personalized learning support • Stress reduces learning efficiency	9. PROBLEM ROOT / CAUSE [RC] • One-size-fits-all learning methods • Emotional state ignored during learning • Lack of personalized academic guidance	7. BEHAVIOR + ITS INTENSITY [BE] • Searches online for solutions frequently • Uses AI tools for study assistance • Often studies under stress
3. TRIGGERS TO ACT [TR] • Poor academic performance • Feeling overwhelmed while studying • Need for instant guidance	10. YOUR SOLUTION [SL] • Detect emotions using BiLSTM and BERT • Generate personalized guidance with Gemini AI • Display analytics through Streamlit • Track emotion history and learning progress	8. CHANNELS OF BEHAVIOR [CH] ONLINE • Learning platforms • AI assistants • Educational websites
4. EMOTIONS BEFORE / AFTER [EM] Before • Stressed • Confused • Frustrated After • Motivated • Confident • Focused		OFFLINE • Classroom • Library • Study groups